Calc 3 – Vector Fields & Line Integrals

Worksheet A

1. Compute $\operatorname{div}\mathbf{F}$ and $\operatorname{curl}\mathbf{F}$ for $\mathbf{F} = \langle x^2, y^2, z^2\rangle$.

2. Compute the line integral $\int_C y\,dx + x\,dy$ where C is the line segment from $(0,0)$ to $(2,3)$.

3. Compute $\int_C x\,ds$ where C is parametrized by $\mathbf{r}(t) = \langle t, t^2\rangle$, $t \in [0,1]$.

4. Show $\mathbf{F} = \langle 2x+y, x+3y^2\rangle$ is conservative and find a potential.

5. Compute the work $\int_C \mathbf{F}\cdot d\mathbf{r}$ for $\mathbf{F} = \langle y, x\rangle$ along $\mathbf{r}(t) = \langle \cos t, \sin t\rangle$, $t \in [0, \pi/2]$.

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Worksheet B

1. Compute $\operatorname{div}\mathbf{F}$ and $\operatorname{curl}\mathbf{F}$ for $\mathbf{F} = \langle yz, xz, xy\rangle$.

2. Compute $\displaystyle\int_C (x+y)\,dx + (y-x)\,dy$ where C is the segment from $(0,0)$ to $(1,1)$.

3. Show that $\mathbf{F} = \langle e^x \sin y, e^x \cos y\rangle$ is conservative; find a potential.

4. Use FTLI: evaluate $\displaystyle\int_C \nabla f \cdot d\mathbf{r}$ where $f(x,y) = x^2 y$ and C goes from $(1,1)$ to $(2,3)$.

5. For $\mathbf{F} = \langle -y, x\rangle$, compute $\displaystyle\oint_C \mathbf{F}\cdot d\mathbf{r}$ on the unit circle (counter-clockwise).

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Answer Key (Worksheets A & B)

Worksheet A.

1. $\operatorname{div}\mathbf{F} = 2x+2y+2z$. $\operatorname{curl}\mathbf{F} = \langle 0,0,0\rangle$ (since each component depends only on its own variable).

2. Parametrize $\mathbf{r}(t) = (2t, 3t)$, $t \in [0,1]$. $dx = 2\,dt$, $dy = 3\,dt$. $\int_0^1 [3t \cdot 2 + 2t \cdot 3]\,dt = \int_0^1 12t\,dt = \boxed{6}$.

3. $\mathbf{r}'(t) = \langle 1, 2t\rangle$, $\|\mathbf{r}'\| = \sqrt{1+4t^2}$. $\int_0^1 t\sqrt{1+4t^2}\,dt$. Let $u = 1+4t^2$, $du = 8t\,dt$. $= \frac{1}{8}\int_1^5 \sqrt{u}\,du = \frac{1}{8} \cdot \frac{2}{3}(5^{3/2} - 1) = \boxed{\frac{5\sqrt{5}-1}{12}}$.

4. $P = 2x + y$, $Q = x + 3y^2$. $P_y = 1 = Q_x$. ✓Integrate $f_x = 2x + y \Rightarrow f = x^2 + xy + g(y)$. $f_y = x + g'(y) = x + 3y^2 \Rightarrow g(y) = y^3$. So $\boxed{f(x,y) = x^2 + xy + y^3 + C}$.

5. $\mathbf{F}$ is conservative ($P_y = 1 = Q_x$). Potential $f = xy$. Endpoints: $\mathbf{r}(0) = (1,0)$, $\mathbf{r}(\pi/2) = (0,1)$. $\int = f(0,1) - f(1,0) = 0 - 0 = \boxed{0}$.

Worksheet B.

1. $\operatorname{div}\mathbf{F} = 0+0+0 = \boxed{0}$. $\operatorname{curl}\mathbf{F} = \langle x-x, y-y, z-z\rangle = \boxed{\mathbf{0}}$. ($\mathbf{F} = \nabla(xyz)$.)

2. Parametrize $\mathbf{r} = (t,t)$, $t \in [0,1]$. $\int_0^1 [(2t)(1) + (0)(1)]\,dt = \int_0^1 2t\,dt = \boxed{1}$.

3. $P_y = e^x \cos y = Q_x$. ✓Integrate $f_x = e^x \sin y \Rightarrow f = e^x \sin y + g(y)$. $f_y = e^x \cos y + g'(y) = e^x \cos y \Rightarrow g = \text{const}$. $\boxed{f = e^x \sin y + C}$.

4. By FTLI: $\int = f(2,3) - f(1,1) = 12 - 1 = \boxed{11}$.

5. Parametrize: $\mathbf{r} = (\cos t, \sin t)$. $\mathbf{F}(\mathbf{r}) = (-\sin t, \cos t)$. $\mathbf{r}' = (-\sin t, \cos t)$. $\mathbf{F}\cdot\mathbf{r}' = \sin^2 t + \cos^2 t = 1$. $\int_0^{2\pi} 1\,dt = \boxed{2\pi}$.